Dynamic Student Registration System

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Introduction

The Dynamic Student Event Registration System is a web application developed to help students register for an event in an easy and interactive way. The system was built using HTML, CSS, JavaScript, and JSON. It allows students to fill in their information based on their academic level, either Undergraduate or Postgraduate, without reloading the page.

The main aim of the project was to create a responsive multi-step registration form with dynamic features and proper validation while maintaining a clean and user-friendly interface.

Technologies Used

HTML

HTML was used to create the structure of the application. It was used to build the form, buttons, modal popup, labels, and input fields.

CSS

CSS was used to style the application and improve the user interface. It helped in creating the modal design, responsive layout, form styling, animations, and overall appearance of the system.

JavaScript

JavaScript was used to add functionality and interactivity to the application. It handled the modal popup behavior, multi-step form navigation, dynamic rendering of fields, form validation, and asynchronous fetching of department data.

JSON

A JSON file was used to store the list of departments. The department data was fetched dynamically into the dropdown menu using JavaScript Fetch API.

Dynamic Rendering Implementation

Dynamic rendering was implemented using JavaScript event listeners and DOM manipulation. When a student selects an academic level, the system automatically displays only the fields related to that level.

For Undergraduate students, the Hostel Name field is displayed while the Last Institution field is hidden. For Postgraduate students, the Last Institution field is displayed while the Hostel Name field is hidden.

This process happens instantly without refreshing the webpage, making the application more interactive and user-friendly.

The form was also divided into two steps. The first step collects personal information, while the second step collects accommodation or institution details. JavaScript was used to switch between the two steps.

Validation Implementation

Validation was added to ensure users enter correct information before submitting the form.

Matric Number Validation

The matric number was validated using JavaScript and Regular Expressions (Regex). Undergraduate students must enter matric numbers beginning with “UG” followed by the current year and four digits, while Postgraduate students must begin with “PG” followed by the current year and four digits.

Date of Birth Validation

The system checks the student’s age using the date of birth entered. Undergraduate students must be younger than 25 years old, while Postgraduate students must be at least 22 years old.

Required Fields Validation

The system also checks that all required fields are filled before the form can be submitted successfully.

Challenges Faced

One of the major challenges faced during the project was implementing dynamic rendering correctly without refreshing the page. This was solved using JavaScript event listeners and CSS class manipulation.

Another challenge was fetching department data from the JSON file. At first, the departments were not loading properly because of file path and fetch issues. This was solved by using async/await with proper error handling.

The modal popup and form scrolling also required adjustments to make the application responsive and accessible on smaller screens.

Conclusion

In conclusion, the Dynamic Student Event Registration System was successfully developed to meet the requirements of the assignment. The project improved understanding of frontend web development concepts such as DOM manipulation, form validation, asynchronous data handling, and responsive design.

The system provides a simple and efficient way for students to register for events while ensuring proper validation and a good user experience.